Design elements for network diagrams

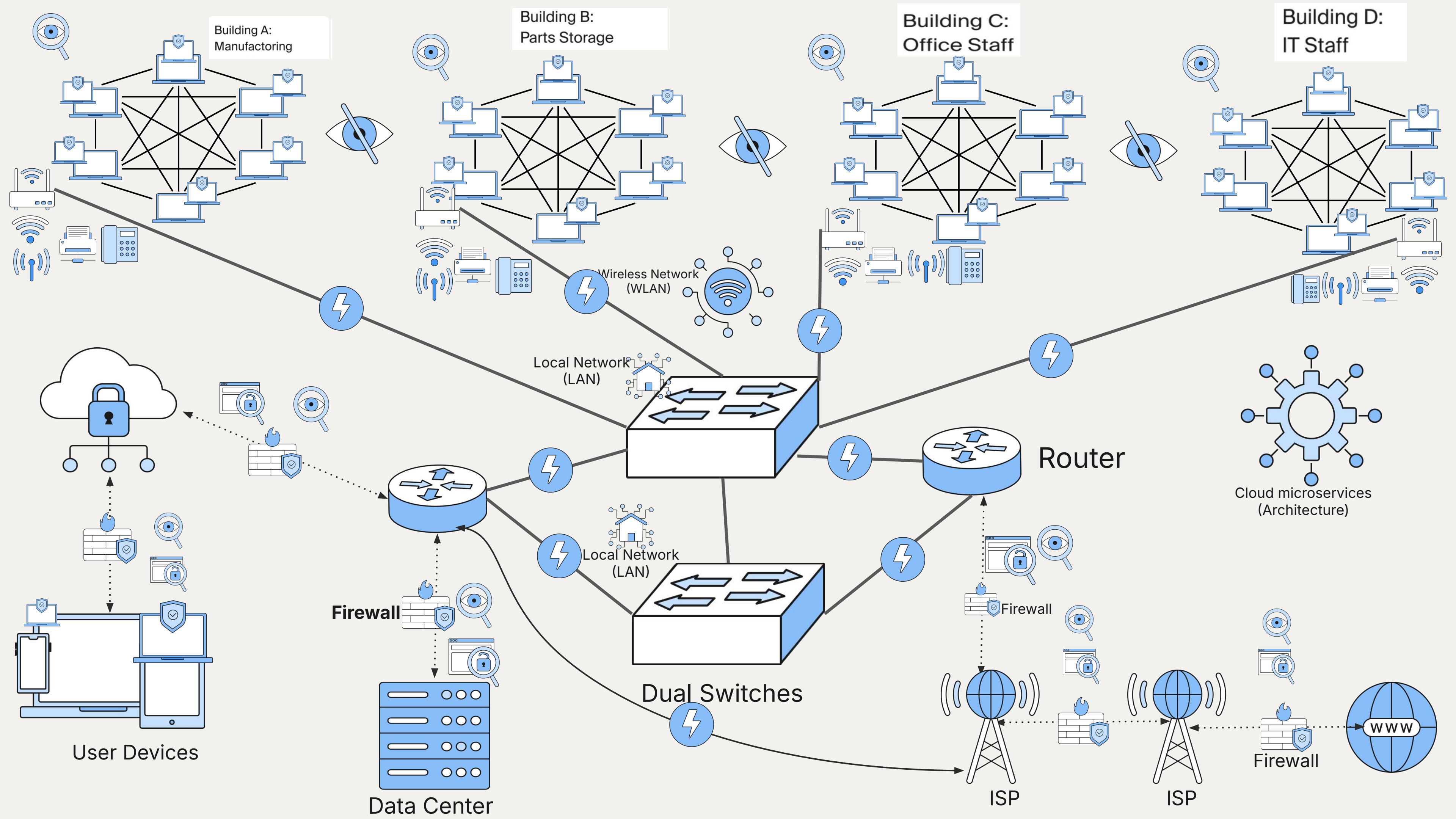
CS Sprint 1 | Final Project

tripleten

Instructions for the template

Do not change this file!

1. Create a copy of this file.
2. Use the elements from this file for your “Tempus Fugit” Project.
3. If you’ve deleted any icons or slides from your own copy, you can easily copy and paste necessary slides from here.
4. When you paste any text from another source, please use the “match destination formatting” option to keep your project’s style consistent.
5. Enjoy!



Building A: Manufactoring
Building B: Parts Storage
Building C: Office Staff
Building D: IT Staff
Wireless Network (WLAN)
Local Network (LAN)
Router
Cloud microservices (Architecture)
Local Network (LAN)
Firewall
Firewall
Dual Switches
User Devices
Data Center
ISP
ISP
Firewall
WWW

REPORT:

A. Mesh topology , and cloud micro services :

    - At the beginning of the client concerns, they indicated the need to modernize also cited that old school is more professional . So I implemented the mesh topology to keep security tight

- Confidentiality is high because wired connections are harder to attack
- It keeps integrity because only trusted sources can access and the information is not tempered with
- And it is readily available because one system down can be isolated and so the rest can continue

B. ISP , Firewall implemented, switches , Routers

        - Also the clients complained of the need for more Internet access and flexibility, meaning they are asking for more availability to keep in touch with other departments while still maintaining a tight cyberspace
        - The introduction of two Internet service provider was a great start because it delivers high speed Internet access to the router which is secured tightly by a customized firewall to detect , defend and recover any cyber attack that may occur
        - Again the addition of two switches instead of one just gives much more room for speed and accuracy and also security because as shown in the project, both routers are connected together, which is connected to the router that delivers local Internet access to the WLAN. One of the router is connected wirelessly to the ISP with firewall security and continuous monitoring in place, while the other one is connected with a high-speed cable to the ISP to serve as a back up in case of a cyber attempts on the wireless network. The wireless network can be easily isolated and defended without hindering Buisness continuity

    - Confidentiality is strong because any access to WAN cross examined with a security control like the firewall in place and switches are connected through high-speed cable, so there is no visibility into the network.
    - Integrity is strong because data arrives in time, secure and correctly.
    - High AVAILABILITY (multiple connections mean no single failure brings down the network)

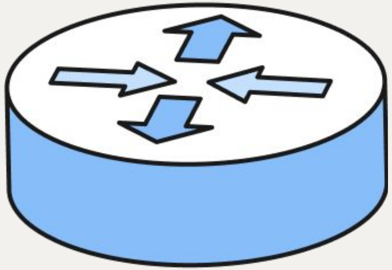
C. WLAN , HIGH SPEED CABELS, REMOTE ACCESS

        - Client expressed the need for flexibility in Intercommunication between departments, so in that case, wireless LAN was implemented for tighter cyber security, but yet improving intercommunication between departments
        - They also complained of slow outdated cables and outages but with the introduction of the WLAN, changing of cables to high-speed cable and mesh topology, it not only improves security but communication
        - Client also expressed the need for remote access which was made possible with the introduction of cloud services and firewalls to serve as security barriers during the transmission of data.
        - Again the IT department complained of lack of knowledge, but that was me to get it also by the introduction of the cloud services which makes management and maintenance easy, secure and reliable

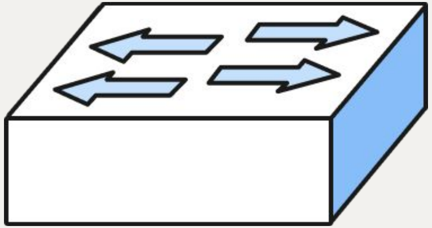
- Confidentiality is strong because of the presence of firewalls at each entry or exit points to the WAN/WWW and also physical cable is much more harder to penetrate since it's on site.
- Again, DATA arrives correctly safely and secure within the WLAN with robust flexibility in communication.
- With the introduction of cloud services, remote access became possible and even more secure due to the firewall barriers in check to ensure that only authorized access is allowed into the company's network. Again, endpoints detection response were installed in all computers and devices that has any form of access either on site or remote in order to constantly monitor, defend and recover data files in case of any attack.

For extra security measures, VPN is recommended to be used to even boost security. SEIM tools are used to monitor, isolate defend and recover data in the event of an attack

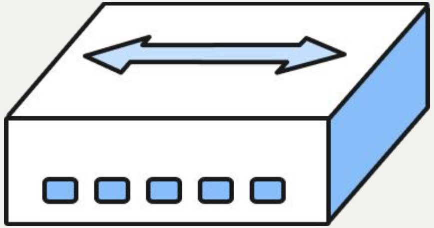
Devices



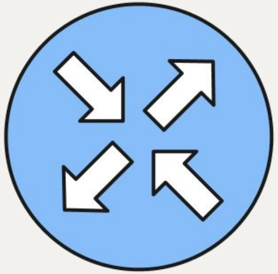
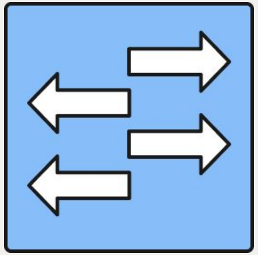
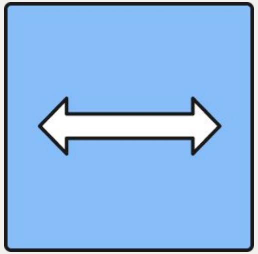
Router



Switch

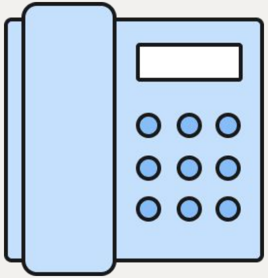


Hub

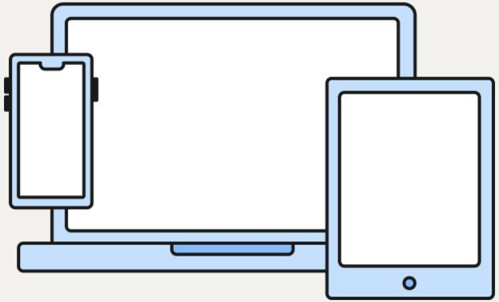




If you need to make a complicated design, use these simplified icons to keep the diagram clean and readable

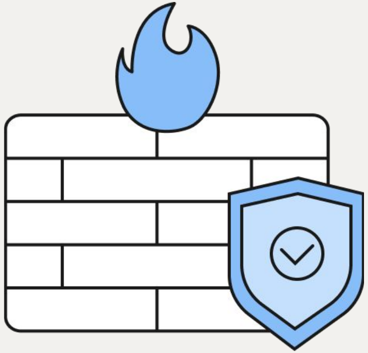
Devices



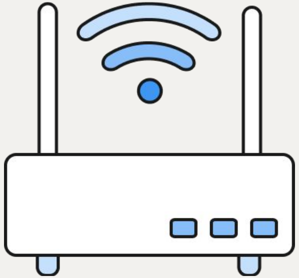
Phone



User Devices



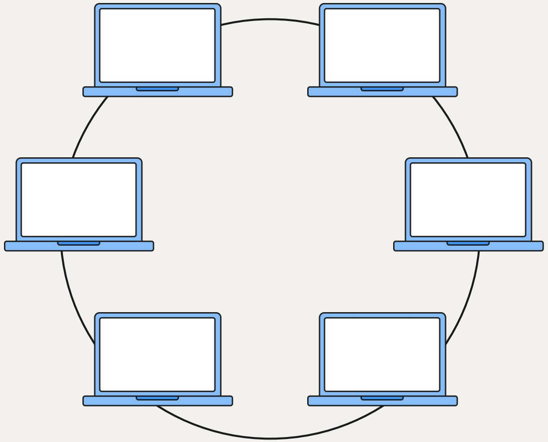
Firewall



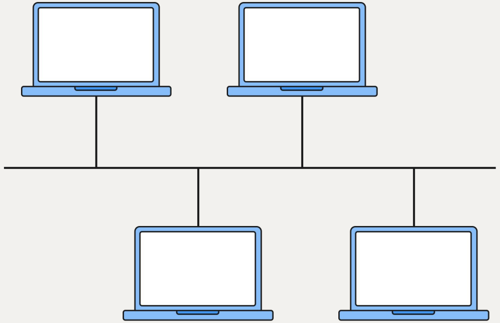
Wi-Fi access point

Printer

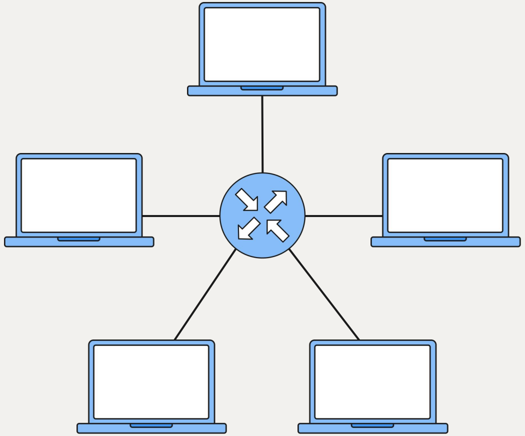
Topologies



Ring topology

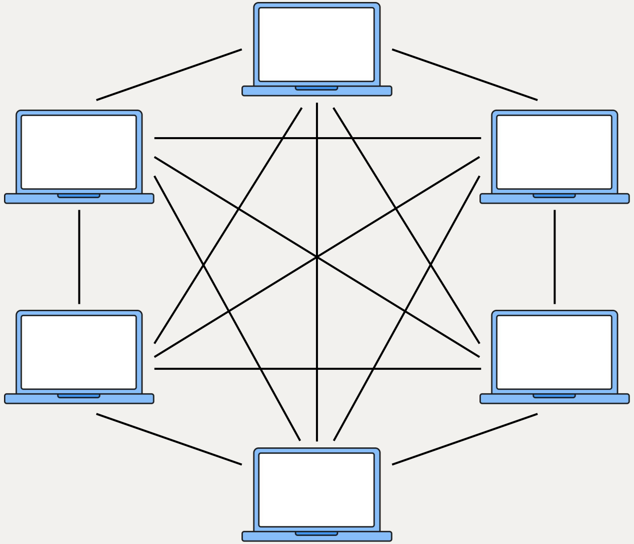


Bus topology

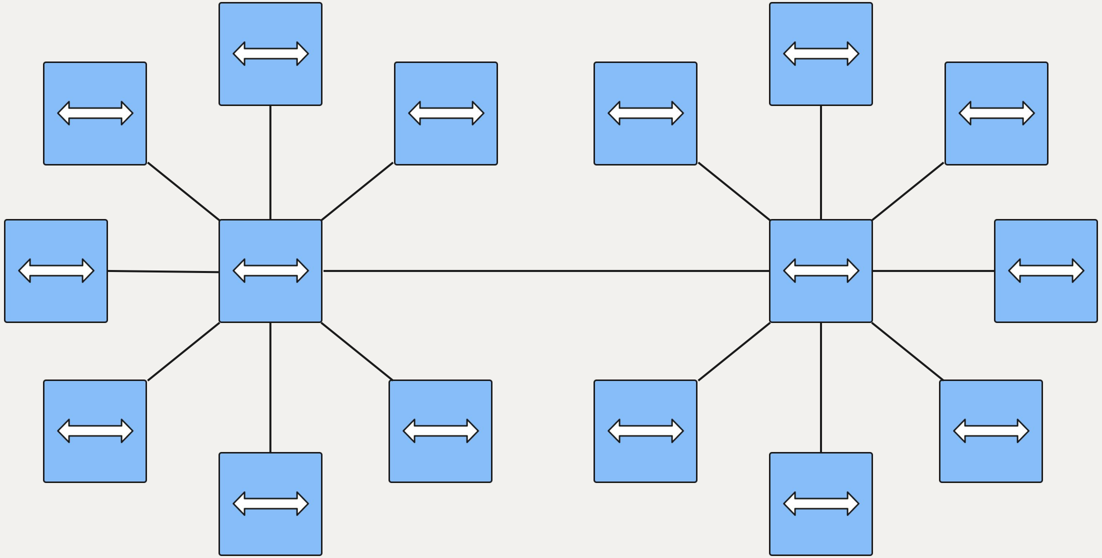


Star topology

Topologies

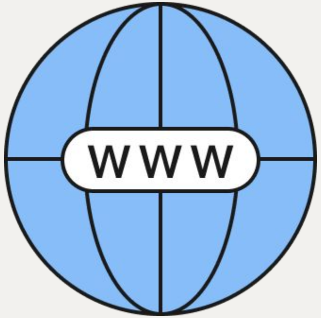


Mesh topology

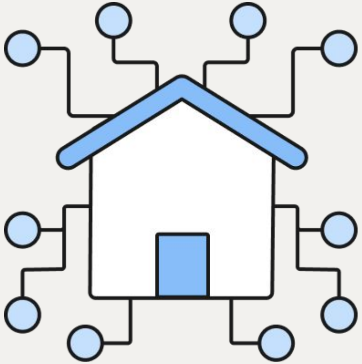


Hub-and-Spoke topology

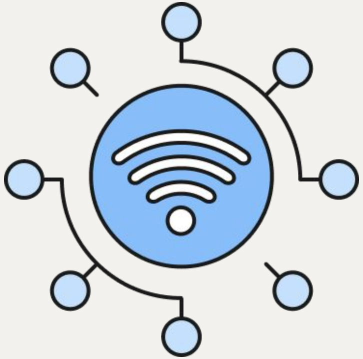
Area Networks



Internet
(WAN)



Local Network
(LAN)

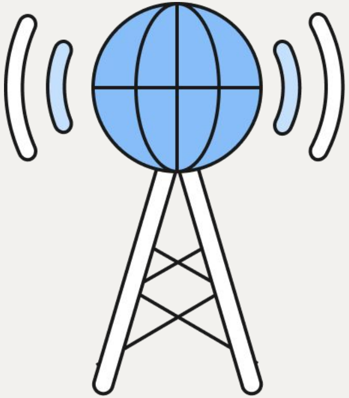


Wireless Network
(WLAN)

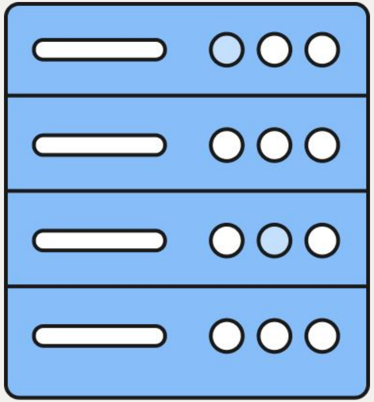
When you need to create connections between the networks, please use the following types of lines:

- Wired network for WAN
- Wired network for LAN
- Wireless network for WLAN

Service Providers



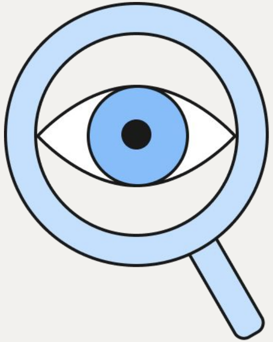
ISP



Data Center

Cloud Service Provider

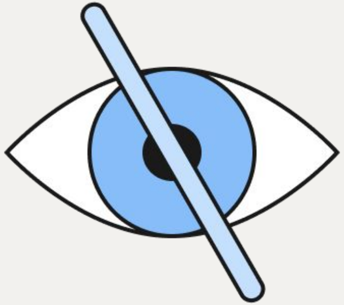
Continuous Monitoring



Network Observability

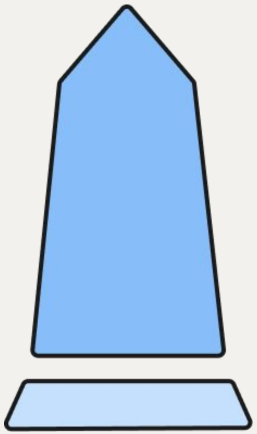
Endpoint Detection & Response (EDR)

Vulnerability Scan

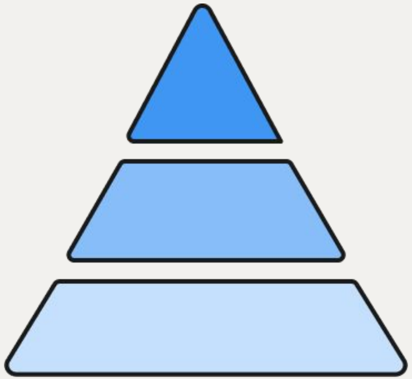


No visibility into network

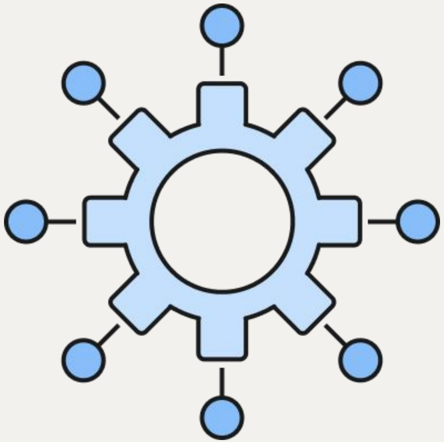
Architectures



Monolithic ERP
(Architecture)



Three-tier
(Architecture)

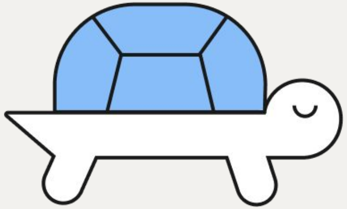


Cloud
microservices
(Architecture)

Connection Types

Wired connections

High-speed Connection



Low-speed Connection

Wireless connections

Wi-Fi Connection

Cellular Connection